JURONG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

CANDIDATE NAME			
CLASS	**18S**	EXAM INDEX	

CHEMISTRY **9729/02**

Higher 2

Paper 2 Structured Questions

29 August 2018
2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
The use of an approved scientific calculator is expected, where appropriate.
A *Data Booklet* is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	**13**
2	**13**
3	**11**
4	**10**
5	**17**
6	**11**
Penalty (delete accordingly)	
Lack **3sf** in final ans	**–1 / NA**
Missing/wrong **units** in final ans	**–1 / NA**
Bond linkages	**–1 / NA**
Total	

This document consists of **20** printed pages.

Answer **all** the questions.

For Examiner's Use

1 Nickel is an important transition metal used in the manufacture of stainless steel alloys. It is first isolated from the mineral ore niccolite by Swedish chemist Axel Fredrik Cronstedt in 1751.

(a) Nickel exists naturally as a mixture of five stable isotopes, each with their own relative isotopic mass. The information about four of these isotopes is given.

isotope	percentage abundance
^{58}Ni	68.1
^{61}Ni	1.14
^{62}Ni	3.63
^{64}Ni	0.93

The relative atomic mass of nickel is 58.7.

Calculate the relative isotopic mass of the fifth isotope of nickel.

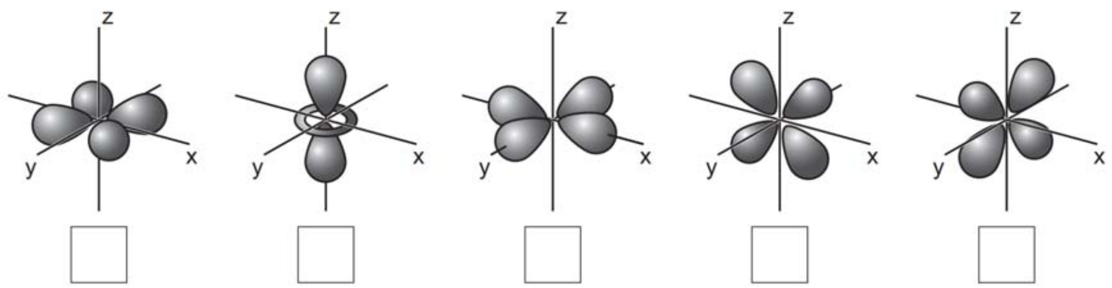
Give your answer to the **nearest whole number**.

[2]

(b) Nickel can form complexes with H_2O ligands.

In a $[Ni(H_2O)_6]^{2+}$ complex, the presence of the H_2O ligands causes the d orbitals to split into two groups at different energy levels.

(i) On the diagram below, tick the box under each of the orbitals of the higher energy level.



[2]

For Examiner's Use

(ii) Explain why a solution of $[Ni(H_2O)_6]^{2+}$ is coloured.

……………………………………………………………………………………..

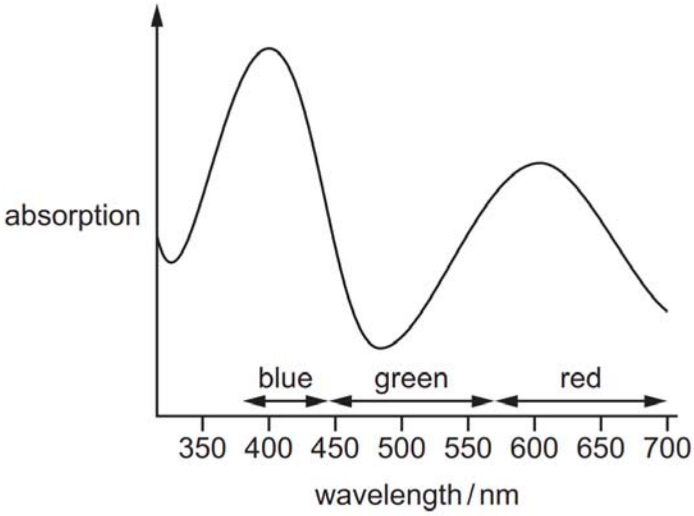
……………………………………………………………………………………..

……………………………………………………………………………………..

……………………………………………………………………………………..

…………………………………………………………………………………….. [2]

(iii) The visible spectrum of a solution of $[Ni(H_2O)_6]^{2+}$ is shown below.



State and explain the colour of the solution.

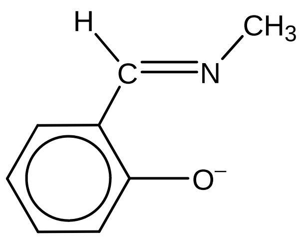
colour of the solution ………………………………..

explanation ………………………………………………………………………….

……………………………………………………………………………………..

…………………………………………………………………………………….. [2]

(c) Nickel can form a complex with ligand **X**$^-$, which has the structure shown below.



ligand **X**$^-$

X$^-$ is a bidentate ligand. On the structure above, draw circles around the atoms that bind to nickel when **X**$^-$ behaves as a ligand. [1]

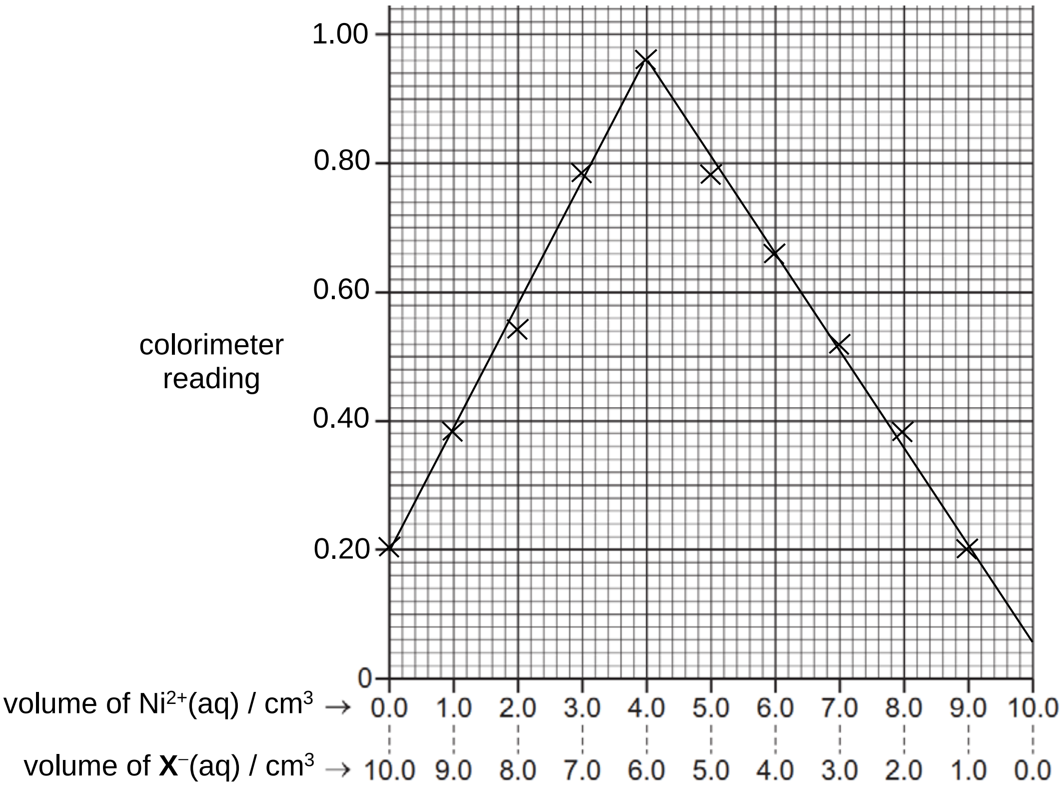
For Examiner's Use

(d) The formula of a complex can be determined using colorimetry.

In colorimetry, light of a certain wavelength is passed through a complex ion solution. The absorbance of the light is proportional to the intensity of the colour of the solution. The more concentrated the complex ion solution, the more intense its colour and so the higher the absorbance.

An experiment is carried out to determine the formula of the complex formed between nickel and ligand **X**$^-$ having the structure given in **(c)**.

The following graph was obtained when the colour intensities of mixtures of a 3×10^{-3} mol dm^{-3} solution of nickel(II) chloride and a 4×10^{-3} mol dm^{-3} solution containing **X**$^-$ were measured using a colorimeter at room temperature.



Determine the stoichiometry of the complex and suggest its structural formula.

[2]

For Examiner's Use

(e) Like nickel, platinum can also forms complexes in which the central atom is surrounded by four ligands.

A and **B** are isomeric complexes of the same shape with the formula of $Pt(PR_3)_2I_2$.
[The PR_3 ligand has the same shape as NH_3. R is a phenyl group.]

Given that isomer **A** has a dipole moment, draw the structures of **A** and **B**.

isomer **A**	isomer **B**

[2]

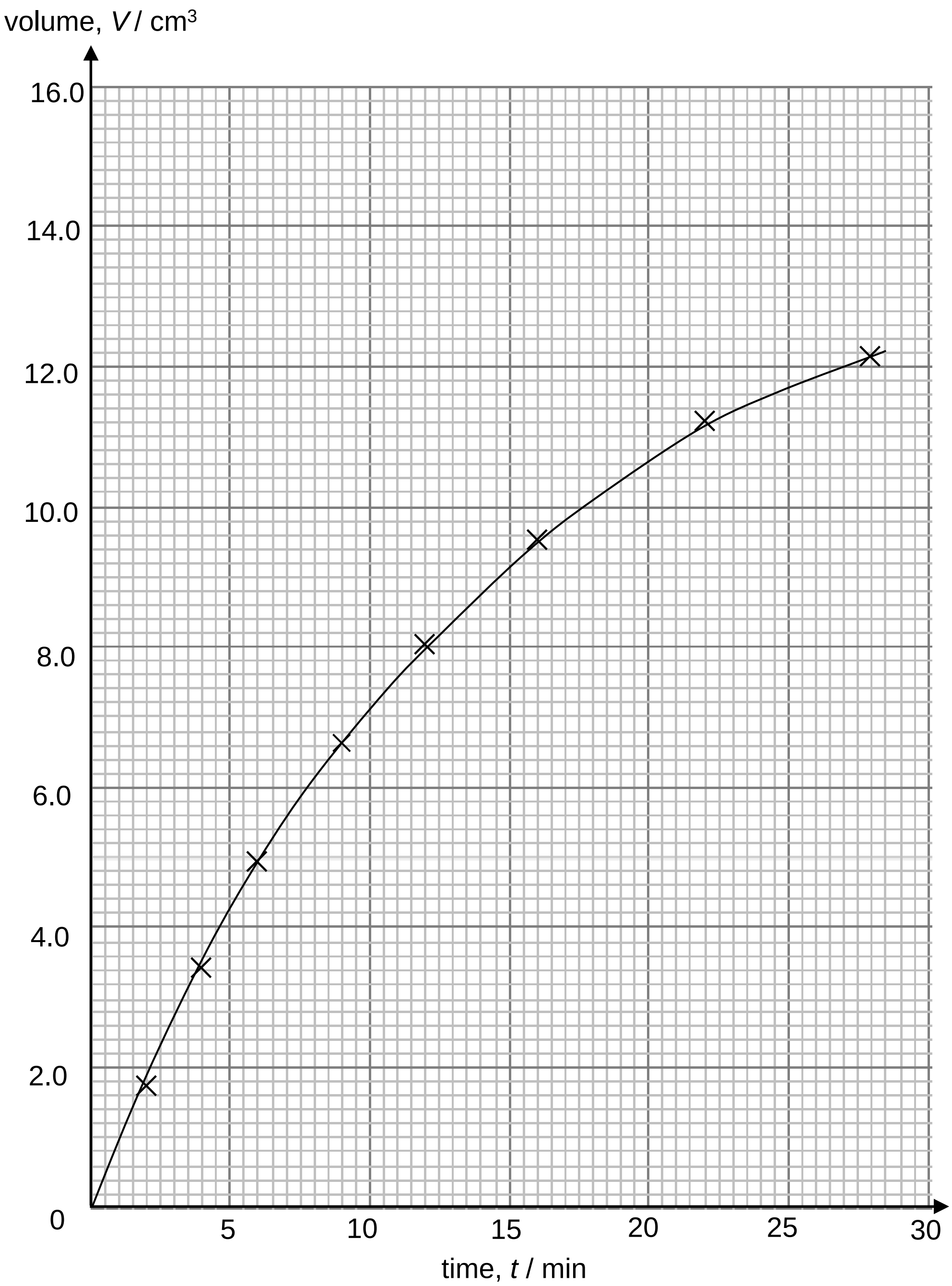
[Total: 13]

2 **(a)** Benzenediazonium chloride, $C_6H_5N_2Cl$, decomposes at 50 °C and 101 kPa, according to the equation below.

$$C_6H_5N_2Cl(aq) \rightarrow C_6H_5Cl(aq) + N_2(g)$$

The progress of the decomposition reaction of a 500 cm^3 sample of $C_6H_5N_2Cl(aq)$ solution was studied by measuring the volume of gas produced over time.

The volume of gas produced, V, after time, t, is proportional to the amount of benzenediazonium chloride that has decomposed. The final volume of gas produced, V_{final}, is 14.6 cm^3 and it is directly proportional to the original concentration of benzenediazonium chloride. The results are recorded in the graph below.



For Examiner's Use

For Examiner's Use

(i) Use the graph to determine the order of the reaction with respect to benzenediazonium chloride.

Show all your working, and draw clearly any construction lines on your graph.

Order of reaction with respect to benzenediazonium chloride: …………….

explanation ………………………………………………………………………….

……………………………………………………………………………………….. [2]

(ii) Calculate the rate constant, stating its units.

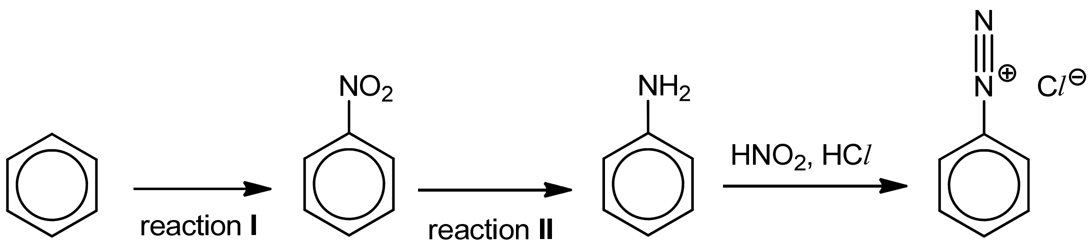
[2]

(iii) Assuming nitrogen behaves ideally, calculate the original concentration, in mol dm^{-3}, of benzenediazonium chloride.

[3]

For Examiner's Use

(b) Benzenediazonium chloride is an important intermediate for the production of dyes. The following scheme shows the synthesis of benzenediazonium chloride from benzene.



benzenediazonium chloride

(i) State the type of reaction and suggest the reagents and conditions for reaction **I**.

type of reaction: ……………………………………………….

reagents: …………………………………………………….

conditions: ………………………………………………….. [3]

In reaction **II**, nitrobenzene is reduced to phenylamine via two steps.

During step 1, granulated Sn and concentrated HC*l* are added to nitrobenzene and the mixture is heated under reflux in a hot water bath for about half an hour. Sn, which acts as the reducing agent, is converted to Sn^{4+} ions.

(ii) Balance the following half-equation for the reduction of nitrobenzene in acid solution to give $C_6H_5NH_3^+$ in step 1.

.... $C_6H_5NO_2$ + ... H^+ + …. e^- $\rightarrow$ ………………………………………….. [1]

(iii) Hence, by considering electron transfer, write a balanced overall equation for the reaction of nitrobenzene, Sn and concentrated HC*l* in step 1.

…. $C_6H_5NO_2$ + …. Sn + …. HC*l* $\rightarrow$ …………………………………………… [1]

Step 2 involves the addition of aqueous sodium hydroxide to the resulting mixture.

(iv) Suggest the purpose of this stage.

……………………………………………………………………………..

……………………………………………………………………………..

…………………………………………………………………………….. [1]

[Total: 13]

For Examiner's Use

3 Metals have properties that make them well suited to serve as battery anodes.

They are easily oxidised from their metallic state to produce ions and electrons, where the electrons liberated are conducted throughout the external circuit. The fact that metals are physically strong and easily fashioned into any desired shape adds to their attractiveness as anodes.

Metals commonly employed as anodes in commercial batteries are listed in **Table 3.1**. The tabulated properties give clues as to the ability of each metal to play this role.

Metal	relative atomic mass	density / g cm^{-3}	standard electrode potential, $E^{\ominus}$ / V	electrochemical capacity / A h g^{-1}
lithium	6.9	0.53	–3.04	3.86
sodium	23.0	0.97	–2.71	1.17
magnesium	24.3	1.74	–2.38	2.21
iron	55.8	7.86	–0.44	0.96
zinc	65.4	7.14	–0.76	0.82
lead	207.2	11.3	–0.13	0.26

Table 3.1

(a) Suggest **two** reasons why lithium is the best choice among the metals in **Table 3.1** to be used as a battery anode.

....

....

....

....

.... [2]

For Examiner's Use

(b) A schematic diagram of a lithium-ion battery is shown below. Lithium is the anode whereas a paste of iron disulfide (FeS_2) powder mixed with powdered graphite serves as the cathode.

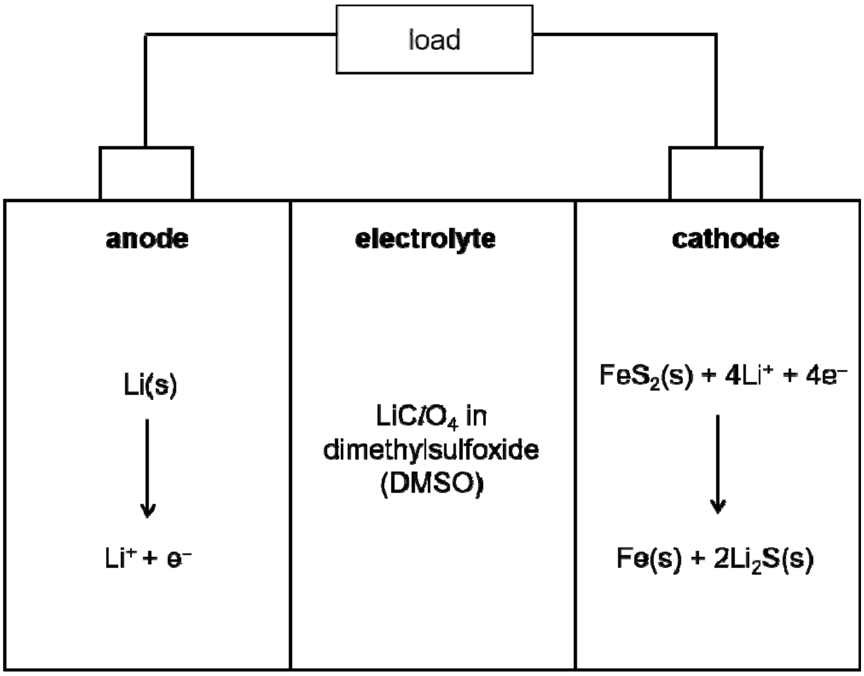


Figure 3.1

(i) On **Figure 3.1**, indicate clearly the direction of electron flow. [1]

(ii) Most batteries use aqueous solutions of ionic salts or ionisable molecules as electrolytes. However, the electrolyte used in this lithium-ion battery must be non-aqueous such as dissolving $LiClO_4$ in dimethylsulfoxide (DMSO), which is an organic solvent. Suggest why.

……………………………………………………………………………………

……………………………………………………………………………………

…………………………………………………………………………………… [1]

For Examiner's Use

(c) Like enthalpy, Gibbs free energy G is a state function. Thus, Hess' Law can likewise be applied to calculate the standard Gibbs free energy change of a reaction, $\Delta G^{\ominus}$, from relevant data such as the standard Gibbs free energy changes of formation.

Table 3.2 below lists the standard Gibbs free energy change of formation, $\Delta G_f^{\ominus}$, for some compounds.

species	$\Delta G_f^{\ominus}$ / kJ mol^{-1}
$FeS_2(s)$	–160
$Li_2S(s)$	–439

Table 3.2

(i) Use the data in **Table 3.2** above to show that the standard Gibbs free energy change, $\Delta G_r^{\ominus}$, of the overall cell reaction in the lithium-ion battery is –718 kJ mol^{-1}.

$$4Li(s) + FeS_2(s) \rightarrow Fe(s) + 2Li_2S(s)$$

[1]

(ii) Use the $\Delta G_r^{\ominus}$ value given in **(c)(i)** to calculate the $E^{\ominus}_{cell}$ of this battery.

[2]

For Examiner's Use

(iii) Use your answer in **(c)(ii)** and relevant $E^{\ominus}$ value in **Table 3.1** to calculate the standard electrode potential of the FeS_2(s)/Fe(s) half-cell at the cathode.

[1]

(iv) By using **one** of the phrases *more positive*, *more negative* or *no change*, deduce the effect of increasing [Li^+] on the electrode potential of

- the left-hand electrode (anode) ……………………………………
- The right-hand electrode (cathode) ………………………………… [2]

(v) Hence deduce whether the overall $E^{\ominus}_{cell}$ is likely to *increase*, *decrease* or *remain the same*, when [Li^+] increases.

……………………………………………………………………………… [1]

[Total: 11]

4 **(a)** **Figure 4.1** is an incomplete sketch showing the melting points of some of the elements of the Period 3 (sodium to argon). Estimate and indicate, on **Figure 4.1**, the melting points of the four other elements: Mg, A*l*, S and C*l*.

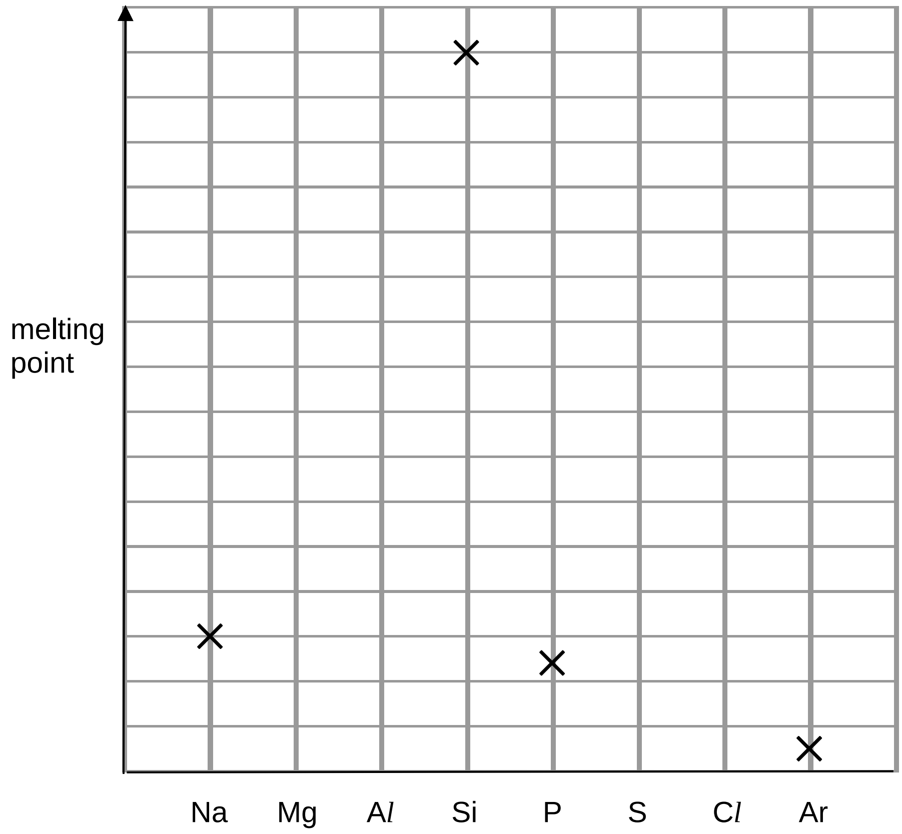
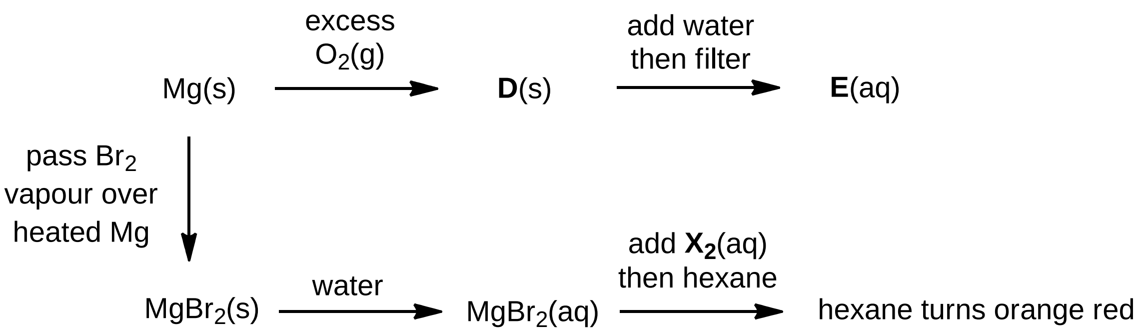


Figure 4.1 [2]

(b) Some reactions of magnesium and its compounds are shown below.



(i) Identify compounds **D** and **E**.

D: ………………….. **E**: ……………………. [2]

(ii) Use appropriate data in the *Data Booklet* to deduce whether $\mathbf{X_2}$ is Cl_2 or I_2.

……………………………………………………………………………………..

……………………………………………………………………………………..

…………………………………………………………………………………….. [2]

(c) Sodium reacts with water to form aqueous sodium hydroxide.

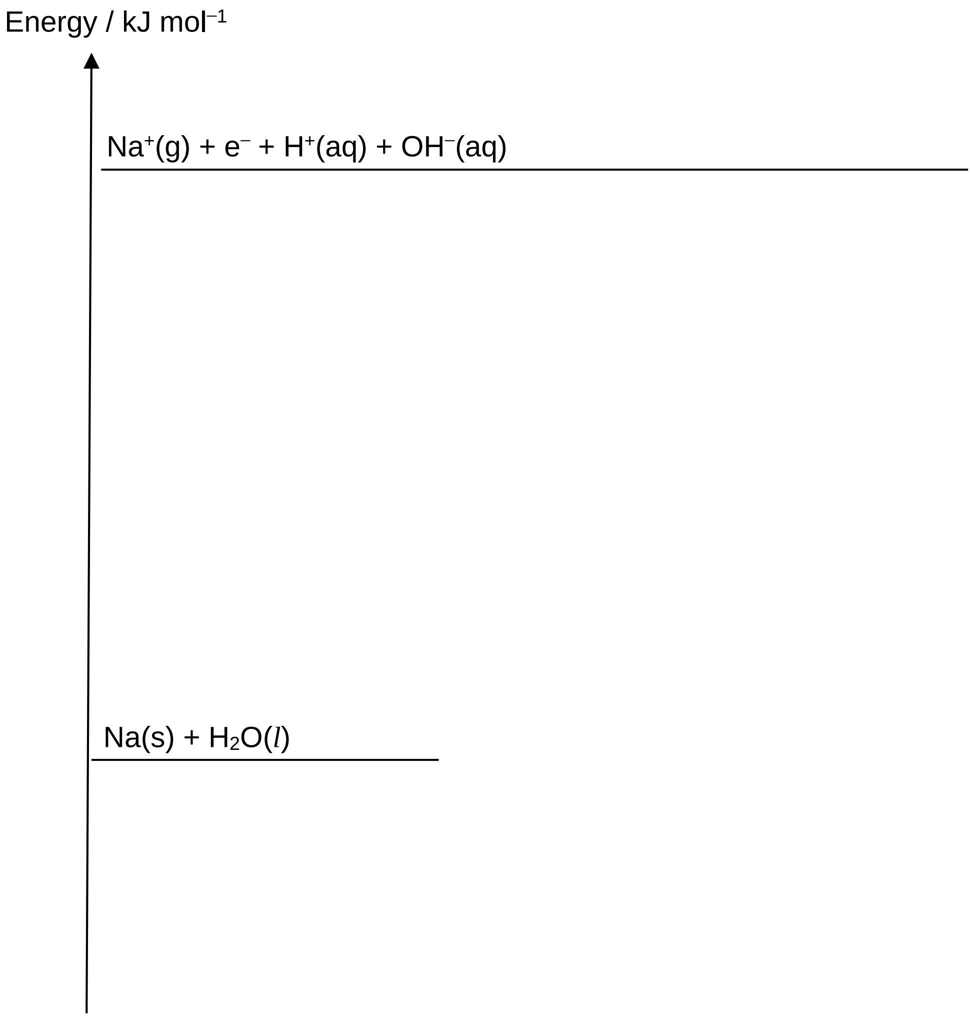
$$Na(s) + H_2O(l) \rightarrow Na^+(aq) + OH^-(aq) + \frac{1}{2}H_2(g) \qquad \Delta \boldsymbol{H}_r$$

On the grid below, draw an energy cycle which can be used to calculate $\Delta \boldsymbol{H}_r$, by incorporating the enthalpy changes in **Table 4.1** and any relevant data from the *Data Booklet*.

Hence, calculate $\Delta \boldsymbol{H}_r$.

	value / kJ mol^{-1}
enthalpy change of atomisation of Na(s)	+107
enthalpy change for $Na^+(g) + H^+(aq) + e^- \rightarrow Na^+(aq) + \frac{1}{2}H_2(g)$	–850
enthalpy change for $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	–58

Table 4.1



[4]

[Total: 10]

For Examiner's Use

5 **(a)** But-1-ene reacts with hydrogen bromide to give 2-bromobutane as the major product.

(i) Name and describe the mechanism for this reaction. Show all charges, relevant lone pairs and the movement of electron pairs by using curly arrows.

Name of mechanism: …………………………………………………

[3]

(ii) With reference to the mechanism you have drawn in **(a)(i)**, explain why the major product is 2-bromobutane rather than 1-bromobutane.

………………………………………………………………………………….

………………………………………………………………………………….

…………………………………………………………………………………. [1]

(iii) 2-bromobutane is chiral. However, the product mixture from this reaction does not rotate plane-polarised light.

With reference to the mechanism you have drawn in **(a)(i)**, explain why this is so.

………………………………………………………………………………….

………………………………………………………………………………….

…………………………………………………………………………………. [1]

(b) **Figure 5.1** shows a reaction scheme involving 4-bromobutanone.

Compounds **J** and **K** have the following properties:

- Effervescence is seen when reacted with sodium metal.
- No yellow precipitate is formed when mixed with alkaline aqueous iodine.
- A pale cream precipitate slowly forms when excess HNO_3(aq) is added, followed by $AgNO_3$(aq).

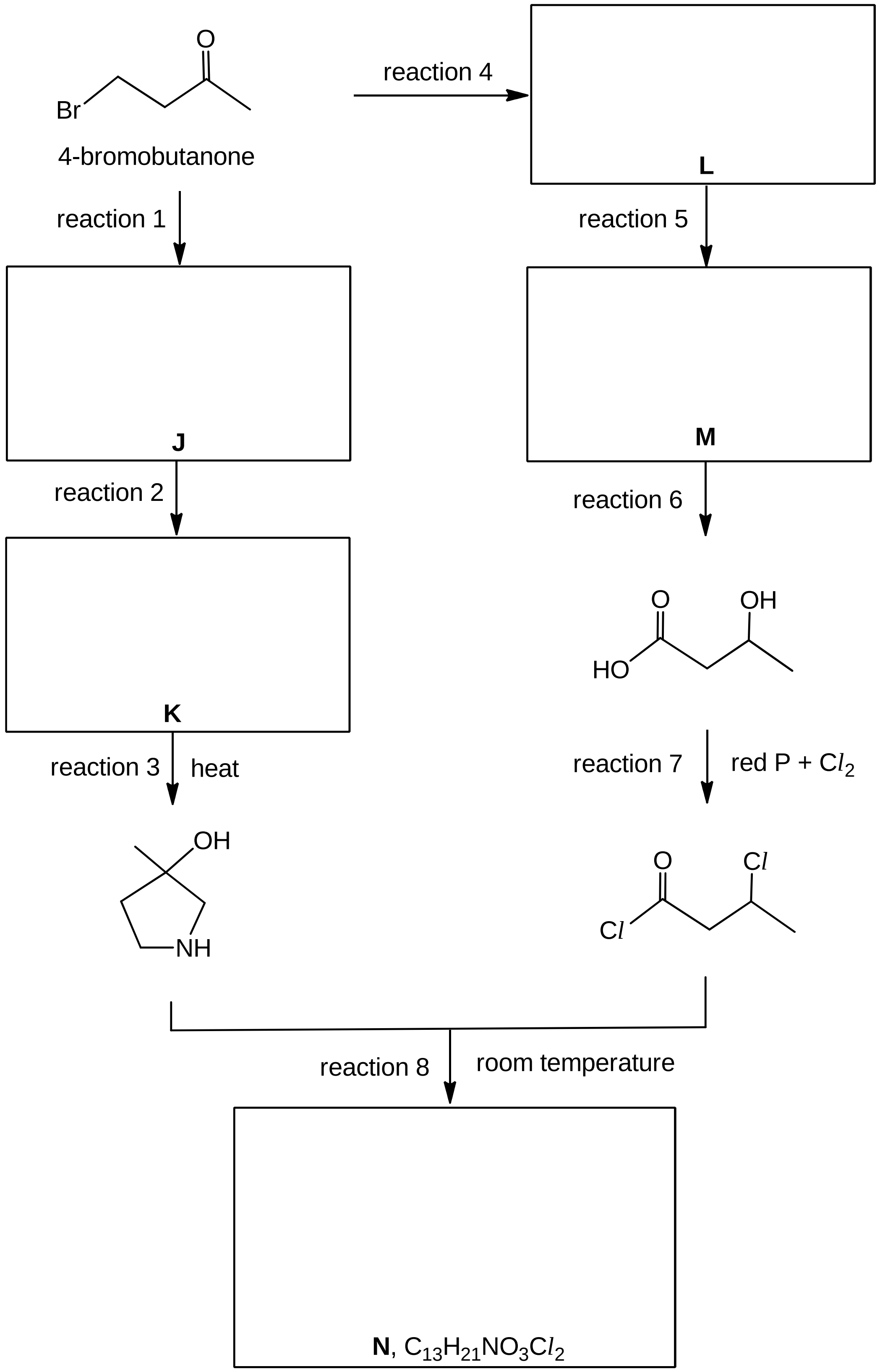


Figure 5.1

For Examiner's Use

(i) Work out the structures for compounds **J**–**N**. Draw their structural formulae in the boxes on the reaction scheme shown in **Figure 5.1**. [5]

(ii) Suggest reagents and conditions for reactions 1, 2, 4, 5 and 6.

reaction 1: …………………………………………………

reaction 2: …………………………………………………

reaction 4: …………………………………………………

reaction 5: …………………………………………………

reaction 6: ………………………………………………… [5]

(iii) State the types of reaction for reactions 7 and 8.

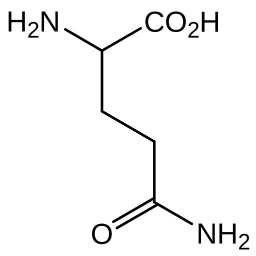
reaction 7: …………………………………………………

reaction 8: ………………………………………………… [2]

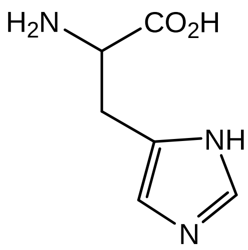
[Total: 17]

For Examiner's Use

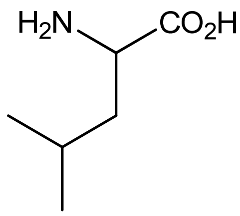
6 A pentapeptide comprises the following five amino acids.



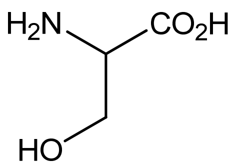
glutamine (gln)
M_r = 146



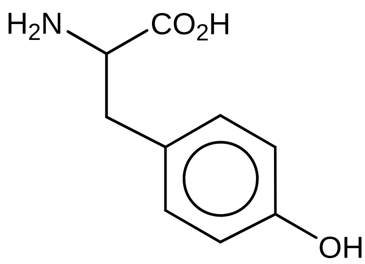
histidine (his)
M_r = 155



leucine (leu)
M_r = 131



serine (ser)
M_r = 105



tyrosine (tyr)
M_r = 181

(a) Calculate the M_r of this pentapeptide. Show your working clearly.

[1]

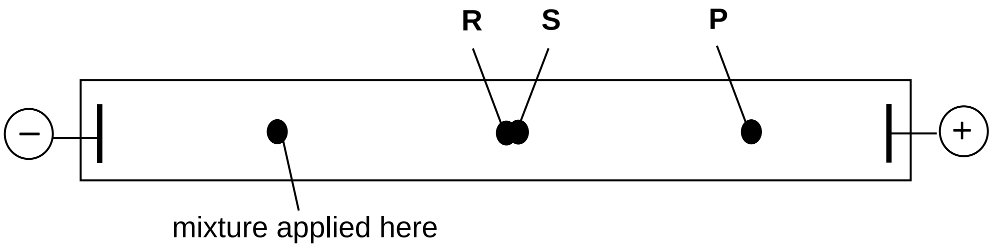
For Examiner's Use

(b) The pentapeptide was broken down by enzymes to form shorter peptides and individual amino acids. One of the peptides is a dipeptide with the sequence **gln-tyr**.

(i) Draw the structure of the dipeptide at pH 12.

[2]

A mixture of this dipeptide (gln-tyr) and its two constituent amino acids (gln and tyr) was subjected to electrophoresis in a buffer at pH = 12. At the end of the experiment, the following results were seen. Spots **R** and **S** remained very close together.



The three spots are due to the three species gln, tyr and gln-tyr.

(ii) Which species is responsible for spot **P**? Explain your answer

spot **P**: …………………….

explanation …………………………………………………………………………

……………………………………………………………………………………..

……………………………………………………………………………………..

…………………………………………………………………………………….. [2]

(iii) Suggest why the other two species give spots **R** and **S** that are so close together.

……………..…………………………………………………………………………

……………………………………………………………………………………..

…………………………………………………………………………………….. [1]

For Examiner's Use

(c) State a reagent you would use and the observations you would make to distinguish tyrosine (tyr) from glutamine (gln).

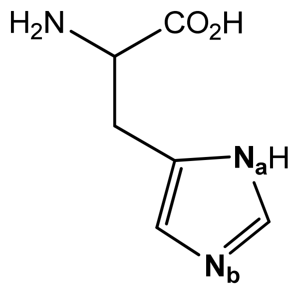
test ……………………………………………

observations ……………………………………………………………………………..

……………………………………………………………………………………………..

……………………………………………………………………………………………… [2]

(d) There are two nitrogen atoms, $\mathbf{N_a}$ and $\mathbf{N_b}$, in the side chain of histidine. However, only one of the nitrogen atoms can act as a Bronsted base.



(i) $\mathbf{N_a}$ and $\mathbf{N_b}$ have the same state of hybridisation. State their state of hybridisation.

……………………………… [1]

(ii) Predict which nitrogen atom, $\mathbf{N_a}$ or $\mathbf{N_b}$, can act as a Bronsted base. Explain your answer.

……………………………………………………………………………………..

……………………………………………………………………………………..

……………………………………………………………………………………..

……………………………………………………………………………………..

…………………………………………………………………………………….. [2]

[Total: 11]

END OF PAPER